TABLE I.- FULL-SCALE DIMENSIONAL CHARACTERISTICS OF THE FULL-SCALE OBSERVATION AIRPLANE

Overall length, ft	40.60
Wing:	
Span, ft	42.00
Area, sq ft	330.00
Mean aerodynamic chord, in.	98.00
Root chord, in.	126.00
Tip chord, in.	63.00
Taper ratio	0.50
Aspect ratio	5.35
Sweep of 0.40 chord line, deg	0
Airfoil section	NACA 2412
Incidence, deg	1.5
Dihedral, deg	6.5
Flap area (total), sq ft	43.6
Deflected-aileron area (total), sq ft	17.5
Aileron area (total), sq ft	22.7
Leading-edge slat area (total), sq ft	41.0

Horizontal tail:	
Area, sq ft	85.0
Span, ft	15.92
Aspect ratio	2.65
Sweep of 0.75 chord line	0
Taper ratio	0.50
Root chord, in.	90.6
Tip chord, in.	45.3
Airfoil section	NACA 0012
Dihedral, deg	6.5
Elevator area (total), sq ft	19.0

Vertical tails:	Inboard	Outboard
Area, sq ft	25.8	21.5
Aspect ratio (one tail)	1.64	2.06
Sweep of 0.55 chord line, deg	0	0
Taper ratio	0.54	0.46
Root chord, in.	61.5	53.0
Tip chord, in.	33.5	24.5
Airfoil section	NACA 0012	NACA 0012
Rudder area, sq ft	11.32	9.5

The model was ballasted to obtain dynamic similarity to the airplane at an altitude of 18,000 feet (ρ = 0.001355 slug/cu ft). Mass characteristics and mass parameters for loadings possible on the airplane and for the loading conditions tested on the model are presented in table II. A remote-control mechanism was installed in the model to actuate the controls for the recovery attempts. Sufficient torque was exerted on the controls to reverse them fully and rapidly for the recovery attempts.

TABLE II.- MASS CHARACTERISTICS AND INERTIA PARAMETERS FOR THE LOADINGS OF THE FULL-SCALE AIRPLANE AND FOR THE LOADINGS TESTED ON THE 1/20-SCALE MODEL

[Values given are full scale, and moments are given about the center of gravity]

Number	Loading	Weight, lb	Center-of-gravity location		Relative density, μ		Moments of inertia, slug-ft^2			Mass parameters		
			x/$\bar{c}$	z/$\bar{c}$	Sea level	18,000 ft	I_X	I_Y	I_Z	$\frac{I_X - I_Y}{mb^2}$	$\frac{I_X - I_Z}{mb^2}$	$\frac{I_Z - I_X}{mb^2}$
Airplane values												
1	Normal	10,423	0.250	-----	9.82	17.24	14,790	21,411	33,579	-116×10^{-4}	-213×10^{-4}	329×10^{-4}
2	Most rearward center of gravity (with full external wing tanks)	11,037	0.304	-----	10.40	18.25	29,346	19,844	47,141	157×10^{-4}	-451×10^{-4}	294×10^{-4}
3	Most forward center of gravity	9,249	0.231	-----	8.71	15.29	12,557	18,853	27,670	-124×10^{-4}	-174×10^{-4}	298×10^{-4}
4	Nominal center of gravity	12,347	0.231	-----	11.63	20.41	13,131	20,687	32,069	-112×10^{-4}	-168×10^{-4}	280×10^{-4}
Model values												
1	Normal	10,602	0.243	0.051	9.98	17.52	21,266	28,777	47,248	-129×10^{-4}	-318×10^{-4}	447×10^{-4}
2	Most rearward center of gravity	10,351	0.290	0.061	9.74	17.09	30,686	24,549	51,985	108×10^{-4}	-485×10^{-4}	376×10^{-4}

The maximum control deflections normally used on the model during the tests (measured perpendicular to the hinge lines) were:

Rudder, deg	25 right, 25 left
Elevator, deg	25 up, 15 down
Ailerons, deg	25 up, 25 down
Flaps:	
Inboard, deg	45 down
Outboard, deg	25 down